Drug Repurposing in Rheumatoid Arthritis

Section 4 — draft after step 11 (Days 1–3)

4.1. Antidiabetic Drugs

Several drugs for type 2 diabetes also act on pathways that drive inflammation in RA, such as AMP-activated protein kinase (AMPK) and NF-κB (Section 3). In RA, insulin resistance also correlates weakly with disease activity ($\rho = 0.21$) and more strongly with inflammatory cytokines such as interleukin-6 ($\rho = 0.63$) [Chung2008]. This section first covers metformin and pioglitazone, followed by glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium–glucose cotransporter-2 (SGLT2) inhibitors.

4.1.1. Metformin

Metformin is a common oral treatment for type 2 diabetes, but how it acts at the cellular level was long unclear [Zhou2001]. The drug appears to act mainly by activating AMPK, which in hepatocytes lowers acetyl-CoA carboxylase activity, increases fatty acid oxidation and suppresses lipogenic enzyme expression [Zhou2001]. Activated AMPK phosphorylates raptor, a step needed to inhibit mammalian target of rapamycin (mTOR) complex 1 (mTORC1) under energy stress [Gwinn2008]. Both mTOR complexes, mTORC1 and mTORC2, promote T helper 17 (Th17) cell differentiation [Nagai2013].

Two animal studies showed that metformin reduced arthritis and protected the joint. In the collagen-induced arthritis (CIA) model (Table 1), immunized mice were given 5 mg/mouse of metformin orally three times per week for 9 weeks, starting 7 days after the first immunization. This treatment significantly reduced arthritis severity and shifted the Th17 to regulatory T (Treg) cell ratio, with fewer Th17 cells and more Treg cells. The numbers of T cells positive for phosphorylated signal transducers and activators of transcription 3 and 5 (STAT3 and STAT5) fell and rose, respectively, and osteoclast activity was reduced [Son2014]. Similarly, in rats with CIA, intraperitoneal (i.p.) metformin (100 mg/kg every 3 days for 3 weeks) protected the knee joints on micro-computed tomography (micro-CT) and on histopathology. Micro-CT showed preserved trabecular bone, histology showed less osteoclast formation and synovitis, and metformin also lowered markers of systemic inflammation, chondrocyte apoptosis and cartilage matrix degradation [Fan2020]. After body-surface-area conversion for a 60-kg adult [NairJacob2016], the oral dose of 5 mg/mouse [Son2014] equals approximately 810–1,220 mg per dose. This range assumes a 20–30-g mouse, because body weight was not reported. The i.p. dose of 100 mg/kg in rats [Fan2020] equals approximately 970 mg per dose. These per-dose values lie within the usual clinical range of 500–2,550 mg/day [Naffaa2020]. Averaged over the dosing intervals of three times weekly and every 3 days, however, they

correspond to approximately 350–520 mg/day and 320 mg/day, respectively, at or below the lower limit of that range [NairJacob2016].

Human joint tissue data also suggest a link between AMPK and RA activity. RA synovium expressed more AMPK-α1 than osteoarthritis synovium; in RA serum, phosphorylated AMPK-α1 correlated weakly with the 28-joint Disease Activity Score (DAS28; r = 0.27), and total AMPK-α1 correlated moderately with tumor necrosis factor-α (TNF-α; r = 0.46) [Chen2020]. In cultured RA fibroblast-like synoviocytes (FLS), metformin inhibited proliferation and migration at millimolar concentrations (5–10 mM) [Chen2020]. It also lowered the levels of TNF-α, interleukin (IL)-1β and IL-6 in these cells [Chen2020].

Three clinical studies suggest a benefit of metformin as an add-on to conventional synthetic (cs) disease-modifying anti-rheumatic drugs (DMARDs). The larger of the two randomized trials, a double-blind study from Egypt, assigned 120 patients with active RA to metformin 1,000 mg or placebo daily, each added to methotrexate (MTX) 7.5 mg/week, over 12 weeks [Abdallah2021]. At 12 weeks, 80.8% of the metformin group reached an American College of Rheumatology 20% improvement (ACR20) response, compared with 54.7% of the placebo group (P = 0.001), without serious adverse effects in either group [Abdallah2021]. A randomized, single-blind trial enrolled 66 Egyptian patients with moderately or highly active RA, of whom

33 received metformin 850 mg twice daily in addition to csDMARDs and 33 received placebo. After 6 months, C-reactive protein (CRP) levels and DAS28-CRP scores were lower than with placebo among the 60 patients who completed the trial, and three metformin-treated patients (9%) withdrew because of gastrointestinal side effects [Gharib2021]. Because these withdrawals were excluded from the analysis, the benefit may be overestimated. In a non-randomized prospective cohort study, 124 patients received low-dose metformin (250 mg twice daily) in addition to traditional csDMARD therapy for 90 days and were compared with 98 patients receiving csDMARDs only. Metformin use was associated with a faster fall in IL-6, TNF-α, high mobility group box 1 (HMGB1) and CRP than in the csDMARD-only group [Zhang2023]. A meta-analysis of four randomized trials with 350 patients found that metformin lowered CRP after 3 months (mean difference −5.99 mg/L, 95% confidence interval [CI] from −7.46 to −4.52) [Alshammari2026]. Over the same period, DAS28 also fell in the three trials that reported it (mean difference −0.80, 95% CI from −0.87 to −0.73) [Alshammari2026]. This interval appears implausibly narrow. The three trials were small, and DAS28 varied by about 1.5 units between patients [Abdallah2021]; the pooled estimate should therefore be read with caution. However, each of the four trials was conducted in Egypt or India, and two carried a high risk of bias because they were single-blind and did not report trial registration [Alshammari2026]. Therefore, metformin may be a

well-tolerated add-on to csDMARDs for lowering RA disease activity in short trials, but larger and longer double-blind randomized controlled trials (RCTs) that also check joint damage on X-ray are needed.

Data on RA prevention are conflicting. In a large UK cohort of people recently diagnosed with type 2 diabetes, metformin use showed no clear association with RA risk compared with other oral diabetes drugs (adjusted hazard ratio [HR] 1.00, 95% CI from 0.82 to 1.22) [Zemedikun2021]. A Mendelian randomization study suggested that metformin may protect against RA (odds ratio [OR] 0.023, 95% CI from 0.0016 to 0.34) [Liang2023]. This effect size is implausibly large for a drug effect and should be interpreted with caution. Genetic variants linked to metformin use may also reflect the genetic liability to diabetes itself. In an Israeli cohort of 113,749 people starting metformin, adherence was associated with a lower RA risk, which was lowest when 40–59% of days were covered rather than under 20% (adjusted HR 0.62, 95% CI from 0.45 to 0.84) [Naffaa2020]. Because the risk was lowest at intermediate rather than the highest adherence, the association did not rise steadily with adherence, which leaves healthy-adherer bias and chance as plausible explanations. In analyses stratified by sex, this association was limited to women [Naffaa2020].

4.1.2. Pioglitazone

Pioglitazone, a thiazolidinedione, activates the nuclear

receptor peroxisome proliferator-activated receptor (PPAR)-γ, and although it is mainly used as an antidiabetic drug, it may also have anti-inflammatory effects. In collagen-induced and serum-transfer arthritis models, PPAR-γ activation by pioglitazone promoted mitophagic flux, which inhibited aberrant activation of the NLR family pyrin domain-containing 3 (NLRP3) inflammasome and reduced the release of IL-1β and IL-18 [Fu2026]. Loss of autophagy, whether genetic or pharmacological, substantially weakened the anti-inflammatory effects of pioglitazone in vitro and in vivo [Fu2026]. Inflammation may also contribute to obesity-induced insulin resistance, although this issue remains debated [Romeo2012].

In animals, pioglitazone was effective only at high doses or when combined with methotrexate. In the adjuvant-induced arthritis (AIA) rat model, oral pioglitazone reduced disease severity, but only at the top dose of 30 mg/kg/day [Koufany2008]. It reduced bone erosions and inflammation-associated bone loss but failed to prevent cartilage lesions, and these anti-arthritic doses were higher than those needed to restore insulin sensitivity and increased body fat [Koufany2008]. Converted with an animal-to-human dose formula, 30 mg/kg/day in rats corresponds to approximately 290 mg/day for a 60-kg adult [Koufany2008][NairJacob2016]. This is more than six times the highest dose given in RA trials, which ranged from 30 to 45 mg/day

[Shahin2011][Ormseth2013][Marder2013]. In a later study using the same model, pioglitazone alone (1.35 mg/kg/day) did not significantly change disease activity [Roy2017]. That dose is equivalent to approximately 13 mg/day for a 60-kg adult [Roy2017][NairJacob2016]. This is below the 30–45 mg/day used in RA trials and may explain the absence of a significant effect of pioglitazone alone. Combined with MTX (0.225 mg/kg/day), it lowered disease activity more than either drug alone ($P < 0.05$) [Roy2017].

Clinical trials show a small benefit at best. In Egypt, a single-blind, nonrandomized trial included 49 patients who had both RA and type 2 diabetes, all maintained on MTX 15 mg/week [Shahin2011]. The 28 patients with more active disease added oral pioglitazone 30 mg/day for 12 weeks, while the other 21 took placebo [Shahin2011]. At 12 weeks, DAS28 ($P = 0.001$) and CRP ($P < 0.0001$) were lower in the pioglitazone group than in the placebo group, and weight gain occurred in 25% of treated patients [Shahin2011]. The placebo group also improved from baseline, and the two groups did not start with the same disease activity, which weakens this result [Shahin2011].

In a US crossover trial that was randomized, double-blind and placebo-controlled, 34 patients with RA on stable therapy received pioglitazone 45 mg/day and placebo for 8 weeks each [Ormseth2013]. Pioglitazone produced a modest 9.3% fall in DAS28-CRP ($P = 0.046$) and no significant change in

DAS28 using erythrocyte sedimentation rate (ESR; P = 0.92), and lower-extremity edema occurred in 16% of participants while on pioglitazone but in none while on placebo [Ormseth2013]. In the same trial, pioglitazone produced a 3.0-mmHg fall in diastolic blood pressure (95% CI −5.7 to −0.2; P = 0.03) [Ormseth2014]. The augmentation index also fell, by 4.7 percentage units (95% CI −7.9 to −1.5; P = 0.004), whereas the reactive hyperemia index (P = 0.46) and pulse wave velocity in the aorta (P = 0.33) did not change significantly [Ormseth2014]. These improvements were independent of pioglitazone's effects on insulin resistance or inflammation, although the P values were not adjusted for multiple comparisons [Ormseth2014].

A larger crossover trial with the same design and dose enrolled 143 non-diabetic patients with RA for 3 months per treatment arm, and DAS28-CRP decreased by a mean of 0.31 on pioglitazone compared with 0.15 on placebo (P = 0.02) [Marder2013]. However, the authors concluded that these findings did not, at that stage, support adding PPAR-γ agonists to the standard treatment of RA [Marder2013]. The benefit in these trials was small and came mainly from CRP-based scores, not from ESR or joint counts [Ormseth2013][Marder2013]. In a nationwide Taiwanese case–control study of patients with type 2 diabetes, including 3,605 patients with RA and 14,420 controls, thiazolidinedione use was associated with lower odds of RA that did not reach significance

(adjusted odds ratio 0.91, 95% CI 0.81 to 1.02) [Hsieh2021]. A higher cumulative dose was also associated with lower RA risk (P for trend = 0.042) [Hsieh2021]. Given the risk of fluid retention and weight gain, pioglitazone should not be used routinely in RA, although it may still be studied as an add-on for patients with RA who also have insulin resistance or type 2 diabetes.

4.1.3. GLP-1 Receptor Agonists

Liraglutide, semaglutide and dulaglutide activate the receptor for glucagon-like peptide-1 (GLP-1) and are therefore called GLP-1 receptor agonists (GLP-1RAs); GLP-1 raises insulin and lowers glucagon secretion in a glucose-dependent manner, slows gastric emptying and reduces food intake [Drucker2018]. Several lines of evidence link these drugs to RA. In a cross-sectional study of 151 patients with RA but without diabetes and 210 sex-matched controls, the patients had an impaired incretin-insulin axis, with higher glucose-dependent insulinotropic polypeptide (GIP) and insulin levels, higher GLP-1 levels after multivariable adjustment and more insulin resistance [TejeraSegura2017]. A review reports that GLP-1 receptors are expressed on circulating T and B lymphocytes in mice and humans and are strongly detectable on human monocytes, but that expression wanes soon after differentiation into macrophages [Bendotti2022]. These findings are not settled, because another review states that most immune

cell subtypes lack functional GLP-1 receptors [Drucker2018]. Immunohistochemical methods for detecting the receptor often lack specificity, and results differ between humans and rodents [Bendotti2022]. In cultured human fibroblast-like synoviocytes from the knee synovium of 15 donors, stimulated with TNF-α (10 ng/mL), dulaglutide (50 or 100 nM, 24 h) lowered expression of IL-1β, IL-6, the chemokine monocyte chemoattractant protein-1 and the alarmin high-mobility group box 1 [Zheng2019]. At the protein level, these cytokines returned close to baseline, although IL-6 mRNA remained about 2-fold above basal levels [Zheng2019]. Dulaglutide also reduced the rise in the degradative enzymes matrix metalloproteinase-3 and -13 [Zheng2019]. In related assays, these effects were accompanied by lower activation of c-Jun N-terminal kinase and by a return of nuclear p65, a nuclear factor kappa B (NF-κB) subunit, to near-baseline levels [Zheng2019]. A review also cites a hypothesis, put forward for osteoarthritis rather than RA, that GLP-1RAs may favor an anti-inflammatory M2 macrophage phenotype [Bendotti2022].

Animal data now add to these cell findings. In AIA rats induced with complete Freund's adjuvant, liraglutide was given at 75 μg/kg/day intraperitoneally from day 15 to day 56 after induction [Nematalla2026]. It inhibited signaling through NF-κB, the NLRP3 inflammasome and IL-1β, as well as through TNF-α and p38 mitogen-activated protein kinase [Nematalla2026]. The

authors report significant therapeutic efficacy in this model, with more pronounced effects when liraglutide was combined with MTX [Nematalla2026]. By body surface area conversion with Km factors of 6 for rats and 37 for humans, this dose corresponds to a human equivalent dose of approximately 12 μg/kg, or about 0.7 mg/day for a 60 kg adult [NairJacob2016]. This lies between the 0.6 mg/day starting dose, which is given only for the first week of titration, and the 1.8 mg/day maximum for type 2 diabetes, and below the 3 mg/day used for weight management [Victoza2025][Saxenda2026]. The route differs, however, as liraglutide was injected intraperitoneally in rats but is given subcutaneously in humans [Nematalla2026][Victoza2025].

Clinical data on GLP-1RAs in patients with RA are scarce, and the main study is a single-center chart review of overweight or obese patients (BMI ≥ 27 kg/m²) [Kellner2025]. It compared 173 patients who took semaglutide (most patients) or tirzepatide with 42 patients who were prescribed one of these drugs but did not take it [Kellner2025]. Disease activity, graded from clinic notes on a 0–3 scale rather than DAS28, changed by −0.03 versus +0.20 in controls ($P = 0.03$), a small difference of unclear clinical meaning [Kellner2025]. Pain fell by 0.6 cm on a visual analogue scale in the treated group but rose by 1.3 cm in controls ($P < 0.001$) [Kellner2025]. ESR and CRP did not fall more than in controls, and the pain relief was not linked to weight

loss, but the study was small and the reasons for starting the drug were not measured [Kellner2025].

Cohort data also link GLP-1RAs to fewer cardiorespiratory and thrombotic events in RA. In adults with RA and obesity but no diabetes, semaglutide or tirzepatide use was linked to fewer first episodes of heart failure (HF) or respiratory failure during days 91–365 than no use (0.7% vs 1.8%; HR of 0.48, 95% CI 0.30 to 0.78) [Loizidis2026]. The authors describe these observations as hypothesis-generating [Loizidis2026]. In a retrospective cohort of 8,697 matched pairs of patients with RA and type 2 diabetes, the 5-year risk of any thromboembolic event was 24% lower with GLP-1 analogs than with DPP-4 inhibitors (HR of 0.76, 95% CI 0.70 to 0.83) [WangQ2025]. Pulmonary embolism alone did not differ significantly (HR of 0.90, 95% CI 0.74 to 1.09), and the authors noted as a limitation that body mass index remained higher in the GLP-1 analog group even after matching [WangQ2025]. In an emulated trial of 1,017 matched pairs of obese adults with type 2 diabetes and RA, semaglutide use was associated with fewer major adverse cardiac and cerebrovascular events than non-GLP-1RA therapy (12.7% vs 16.5%; HR of 0.75, 95% CI 0.60 to 0.94) [Malik2026]. Comparators included DPP-4 inhibitors, SGLT2 inhibitors, sulfonylureas and thiazolidinediones [Malik2026]. Incident HF was also lower with semaglutide (8.9% vs 12.5%; HR of 0.69, 95% CI 0.53 to 0.91), whereas all-cause mortality, myocardial

infarction and stroke did not differ significantly [Malik2026]. The authors attribute the overall benefit mainly to this lower HF risk [Malik2026]. The authors state that the observational design limits causal inference [Malik2026].

Not all data indicate a benefit. A case-control study compared 4,535 RA patients with 22,675 controls and found that use of semaglutide or liraglutide in the 10 years before diagnosis was linked to higher odds of new-onset RA (adjusted odds ratios 1.64 and 1.58) [Israel2026]. Dulaglutide showed a similar but non-significant trend (adjusted odds ratio 1.31) [Israel2026]. This link was seen mainly with short use (≤6 months), which the authors read as a possible transient pro-inflammatory effect early in treatment, although they note that causality cannot be inferred [Israel2026]. However, part of this may reflect confounding by indication or reverse causality, because obesity and diabetes, the main reasons for giving these drugs, are independent risk factors for RA [Israel2026]. The authors add that sustained use may offset this effect through weight loss and better insulin sensitivity [Israel2026]. This case-control study addressed new-onset RA, whereas the chart review above included patients with established RA; the two results are therefore not necessarily in conflict [Israel2026][Kellner2025].

To date, the evidence is limited, observational and mixed, and we found no randomized trials in RA. The

rat study and the cohort data on heart failure, respiratory failure and thrombotic events were favorable, but all of them are preclinical or observational. GLP-1RAs may help some patients with RA and obesity, but they cannot be recommended yet. A possible cardiovascular benefit may be an additional reason for trials, which should also check whether the first months of treatment briefly worsen inflammation.

4.1.4. SGLT2 Inhibitors

SGLT2 inhibitors, such as dapagliflozin, are oral drugs for type 2 diabetes that block glucose reabsorption in the kidney [ElDemerdash2025]. The only animal study in arthritis that we found reported clear anti-inflammatory effects. In an Egyptian study (Ain Shams University) of AIA rats induced with complete Freund's adjuvant, three oral daily doses of dapagliflozin (1, 5 and 10 mg/kg) were tested for 3 weeks; the highest dose worked best [ElDemerdash2025]. Converted by body surface area, with Km factors of 6 for rats and 37 for humans, the highest dose in these male Sprague-Dawley rats corresponds to approximately 97 mg/day for a 60 kg adult [ElDemerdash2025][NairJacob2016]. This is about ten times the 10 mg once-daily dose in the dapagliflozin label, whereas the lowest rat dose corresponds to approximately 10 mg/day, close to the label dose [Farxiga2026]. Compared with untreated arthritic rats, dapagliflozin alone lowered serum NF-κB

p65 by 66% and TNF-α by 71% [ElDemerdash2025]. It raised the p-AMPK/t-AMPK ratio in the joint by 374%, whereas MTX alone did not change it, which suggests involvement of the AMPK pathway described for metformin (Section 4.1.1) [ElDemerdash2025]. It also raised joint gene expression of the autophagy markers autophagy-related gene 7 (ATG-7), Beclin-1 and unc-51-like kinase 1 (ULK-1) [ElDemerdash2025]. On X-ray, neither drug alone reduced bone erosion or joint-space narrowing; only the two drugs together did [ElDemerdash2025].

Human data on SGLT2 inhibitors and RA onset are scarce; one study used TriNetX records to emulate target trials in adults with newly diagnosed type 2 diabetes [Yen2026]. It compared SGLT2 inhibitors with sulfonylureas, dipeptidyl peptidase-4 (DPP-4) inhibitors or pioglitazone (more than 80,000 matched pairs per comparison) [Yen2026]. In the first year, SGLT2 inhibitor use was linked to a slightly lower risk of new RA than sulfonylurea use (0.118% vs 0.131%; HR 0.90, 95% CI 0.84–0.97), whereas the differences against DPP-4 inhibitors (HR 0.91) and pioglitazone (HR 0.92) were not significant [Yen2026]. The gap with sulfonylureas shrank with longer follow-up and lost significance, as it also did in a per-protocol analysis [Yen2026]. The authors suggest that one year may be too short, because RA has a long silent phase before diagnosis [Yen2026].

Unlike GLP-1RAs (Section 4.1.3), SGLT2 inhibitors

were not linked to a higher RA risk than other glucose-lowering drugs [Yen2026]. Clinical data in patients who already have RA are scarce and inconclusive. In a cohort of 277 matched pairs of adults with RA and type 2 diabetes, SGLT2 inhibitor use was associated with lower odds of major adverse cardiovascular events than DPP-4 inhibitor use, but not significantly so (odds ratio of 0.792, 95% CI 0.567 to 1.107) [Kachhadia2026]. The absolute risk difference was −5.8 percentage points (P = 0.172), post-hoc power was approximately 40–50%, and follow-up differed markedly between groups (medians of 3.1 and 6.4 years) [Kachhadia2026]. The authors consider the findings hypothesis-generating [Kachhadia2026]. A 2024 review noted that clinical evidence on SGLT2 inhibitors in rheumatic diseases is limited, and the articles it covered dealt with systemic lupus erythematosus, gout and lupus nephritis [Nallathambi2024]. In Egypt, two randomized trials of SGLT2 inhibitors added to MTX in patients with RA have been registered, both with DAS28-CRP as the primary outcome, but neither had started recruiting at the time of writing [Tanta2024][AlAzhar2026]. Their use in RA therefore remains experimental.

4.1.5. DPP-4 Inhibitors

In diabetes cohorts, DPP-4 inhibitors have served mainly as the comparator group rather than as the treatment under study. They were the control arm for

GLP-1RAs in 8,697 matched patients per group [WangQ2025] and in 4,607 propensity score-matched patients per group [Yu2026]. Similarly, they were compared with SGLT2 inhibitors in 284 versus 823 patients before matching [Kachhadia2026] and in 206,069 matched pairs [Yen2026], and they formed part of the comparator group in a semaglutide cohort [Malik2026]. In RA with type 2 diabetes, GLP-1RAs were associated with lower all-cause mortality than DPP-4 inhibitors (HR 0.68, 95% CI 0.58 to 0.80) [Yu2026]. Major cardiovascular events, however, did not differ between the two groups (HR 0.96, 95% CI 0.89 to 1.04) [Yu2026]. In the primary analysis of one population-based cohort, initiating a DPP-4 inhibitor was linked to a modestly lower RA risk than metformin plus another non-DPP-4 agent (HR 0.66, 95% CI 0.44 to 0.99) [Kim2015]. However, when thiazolidinedione or sulfonylurea initiators served as the reference, the lower risk did not reach significance (HRs 1.04 [0.51–2.12] and 0.66 [0.38–1.15], respectively) [Kim2015]. The authors also noted that residual confounding could not be excluded [Kim2015]. By contrast, a nested case-control study of diabetic patients (790 RA cases, 11,850 controls) found no significant association with RA after adjustment (adjusted OR 1.16, 95% CI 0.94 to 1.43) [Kathe2018]. Only the non-insulin subgroup showed a positive association (OR 1.30, 95% CI 1.001 to 1.985), which is at most a signal because the lower limit barely exceeds 1 [Kathe2018].

The laboratory data we found are limited and conflicting. In an Egyptian study conducted in Alexandria, oral sitagliptin (100 mg/kg/day) reduced hind paw swelling in AIA rats compared with untreated animals [Ibrahim2020]. Moreover, sitagliptin lowered serum TNF-α, IL-6, anti-cyclic citrullinated peptide (anti-CCP) antibodies and receptor activator of NF-κB ligand (RANKL), a driver of osteoclast formation, relative to the untreated arthritic group ($P < 0.001$) [Ibrahim2020]. Its effect on the clinical arthrogram score, which was assessed by examination rather than imaging, and on synovial phosphorylated STAT3 (p-STAT3) protein was smaller than that of tofacitinib, which was tested in parallel [Ibrahim2020]. A contrasting in vitro report found lower DPP-4 levels in RA than in normal synovial tissue, and sitagliptin and vildagliptin raised the release of proinflammatory cytokines from RA synovial fibroblasts [Han2021]. The authors inferred a possible increase in RA risk from these cell experiments [Han2021]. The evidence therefore rests on conflicting observational studies and on animal and in vitro work, and we found no randomized trial of DPP-4 inhibitors in RA in the registries searched.

4.2. Lipid-Lowering Drugs

Patients with RA have up to twice the risk of atherosclerotic cardiovascular disease seen in the general population, a level similar to that in diabetes

[Hansildaar2021], which makes lipid-lowering drugs a logical choice to test. The two classes examined here, statins and fenofibrate, also affect immune cells and may benefit both the joints and the heart (Sections 4.2.1 and 4.2.2).

4.2.1. Statins

Statins act mainly as competitive inhibitors of 3-hydroxy-3-methylglutaryl-CoA reductase (HMG-CoA reductase), binding the enzyme at nanomolar concentrations and displacing its natural substrate, which blocks de novo cholesterol synthesis [Greenwood2006]. Statins also have effects beyond lipid lowering, because fewer isoprenoid intermediates leave small guanosine triphosphatases (GTPases) such as Ras, Rho, Rac and Cdc42 less prenylated and therefore less active [Greenwood2006]. They also block interferon-γ (IFN-γ)-induced major histocompatibility complex (MHC) class II expression and thus MHC-dependent T-cell activation [Kwak2000], and T cells form the largest lymphocyte population in rheumatoid synovium [Gao2024].

Animal data on statins are conflicting. In a preclinical study, mice with serum-induced arthritis were given atorvastatin, pravastatin or simvastatin intravenously (0.2 mg/kg on days 3, 5 and 7), and atorvastatin and pravastatin lessened clinical arthritis and preserved joint structure on paw histology, but simvastatin had no significant effect [Walker2017]. This regimen

corresponds to approximately 1 mg per dose in a 60-kg adult after body-surface-area conversion [NairJacob2016], far below the 40 mg/day used in RA trials [McCarey2004][Kitas2019], although intravenous and oral doses are not directly comparable.

An earlier CIA study with simvastatin was favorable. In mice given daily i.p. simvastatin at 10, 20 or 40 mg/kg between days 21 and 40, arthritis was suppressed in a dose-dependent manner [Leung2003], and a 14-day course of 40 mg/kg, started one day after clinical onset, also inhibited established disease [Leung2003]. Because these doses did not significantly lower plasma cholesterol, the effect appeared largely independent of lipid lowering [Leung2003]. Findings with other statins were less favorable. In a later CIA study, atorvastatin and rosuvastatin had no effect on clinical, histologic, immunologic or biochemical markers of disease [Palmer2004]. The benefit of i.p. simvastatin at 40 mg/kg/day, the dose used by Leung et al., was reproduced, but the authors attributed it to severe toxicity with raised glucocorticoid levels [Palmer2004]. Deaths reduced this group from 18 to 7 mice over the study, which suggests that the earlier benefit may reflect toxicity rather than a true antiarthritic effect [Palmer2004]. When Vandebriel et al. gave CIA mice oral atorvastatin at 0.2 mg/day or oral pravastatin at 0.8 mg/day, arthritis started earlier, and every treated mouse became arthritic, against 7 of 12 controls [Vandebriel2012]. After body-surface-area conversion

for a 60-kg adult [NairJacob2016], the i.p. simvastatin doses of 10–40 mg/kg [Leung2003] correspond to approximately 49–195 mg/day, with the highest dose about five times the 40 mg/day used in RA trials [McCarey2004][Kitas2019]. The oral atorvastatin dose used by Vandebriel et al., about 10 mg/kg, corresponds to approximately 49 mg/day and is close to the trial dose, whereas pravastatin at four times that dose corresponds to approximately 195 mg/day [Vandebriel2012]. Earlier arthritis onset after statin treatment has, to our knowledge, been reported only by this group [Vandebriel2012].

Results in patients are mixed. Two meta-analyses showed that statins reduced disease activity measured by DAS28, with standardized mean differences (SMDs) of −0.55 and −0.79 (95% CIs −0.83 to −0.26 and −1.08 to −0.49, respectively) [Xing2015][Ren2023]. However, the included trials were small and mostly conducted at single centers [Xing2015], and both meta-analyses reported high heterogeneity [Xing2015][Ren2023]. The larger one also found possible publication bias for DAS28 (Egger test P = 0.012) [Ren2023]. In an unblinded randomized study conducted in Egypt, 30 patients with early RA were divided into two groups of 15 and treated for 6 months with MTX (0.2 mg/kg/week) and prednisone (10 mg/day), either alone (group 1) or with atorvastatin 40 mg/day (group 2). DAS28 was significantly reduced from 6.09 to 3.9 in group 2 compared with 6.19 to 5.27 in group 1

[ElBarbary2010]. The double-blind Trial of Atorvastatin in Rheumatoid Arthritis (TARA) randomized 116 patients to atorvastatin 40 mg/day or placebo added to their existing DMARDs [McCarey2004], with co-primary outcomes of change in DAS28 and European League Against Rheumatism (EULAR) response [McCarey2004]. After 6 months, DAS28 improved more with atorvastatin than with placebo (between-group difference −0.52, P = 0.004; 95% CI from −0.87 to −0.17) [McCarey2004]. This difference was slightly below the 0.6-unit change the trial was powered to detect, but EULAR response was more frequent with atorvastatin (31% vs 10%) [McCarey2004], whereas the small unblinded Egyptian study reported a between-group difference of about 1.3 units [ElBarbary2010]. In a 12-week randomized study without placebo (blinding not reported), 55 patients with RA received atorvastatin 20 mg/day or no statin [Tang2011]. DAS28 fell by 0.41 with atorvastatin and by 0.03 without it (P < 0.001), and CRP and ESR also decreased [Tang2011]. The DAS28 change, however, was below the 0.6-unit threshold used to power TARA [McCarey2004]. In contrast, the largest trial we found, the double-blind Trial of Atorvastatin for primary prevention of Cardiovascular Events in RA (TRACE RA), enrolled 3,002 UK patients for cardiovascular end points [Kitas2019]. It found no difference in DAS28 between the atorvastatin and placebo groups [Kitas2019]. Atorvastatin did lower CRP, but the authors judged the fall too small to matter clinically

[Kitas2019]. TRACE RA targeted cardiovascular end points, reported compliance in its atorvastatin arm fell from 89% at 3 months to 39% by 60 months, and the trial was stopped early because cardiovascular events were fewer than expected [Kitas2019]. These design features, rather than a lack of efficacy alone, may partly explain why TRACE RA did not confirm the smaller trials.

Observational data on statin use and RA onset are inconsistent: in an Israeli cohort of 211,627 new statin users, persistence was measured using the proportion of days covered (PDC) [Chodick2010]. Compared with a PDC of <20%, a PDC of ≥80% was associated with lower RA risk (adjusted HR 0.58, 95% CI from 0.52 to 0.65), and to test for healthy-user bias, the same study also examined osteoarthritis [Chodick2010]. For osteoarthritis, high persistence was associated with only a modest risk reduction (HR 0.85, 95% CI from 0.81 to 0.88) [Chodick2010]. A US case–control study that matched 16,363 RA cases with the same number of controls initially found a slightly higher risk with statin use [Peterson2021], but after adjustment for hyperlipidemia the association disappeared, with OR 0.95 (95% CI from 0.90 to 1.01) [Peterson2021]. However, hyperlipidemia is often diagnosed when statin treatment begins, which may limit the accuracy of this adjustment [Peterson2021]. Genetic data, in contrast, suggest a protective role. In a drug-target Mendelian randomization study, genetically proxied HMG-CoA

reductase inhibition was linked to lower RA risk in both the discovery (OR 0.45, 95% CI from 0.32 to 0.63) and replication (OR 0.37, 95% CI from 0.23 to 0.61) datasets [Hu2024]. Although consistent, these data come from European populations only and address RA onset rather than disease activity in established RA [Hu2024]. A small prevention trial was also inconclusive: among 62 anti-citrullinated protein antibody (ACPA)-positive patients with arthralgia, atorvastatin 40 mg/day was compared with placebo for a planned 3 years, but the trial was stopped early because of low inclusion [vanBoheemen2021]. Arthritis developed in 9/31 (29%) atorvastatin recipients and 6/31 (19%) placebo recipients (HR 1.40, 95% CI from 0.50 to 3.95), a non-significant difference [vanBoheemen2021]. Only 62 of the planned 220 participants were enrolled, which left the trial too small to confirm or exclude a protective effect [vanBoheemen2021]. For now, statins in RA are prescribed to lower lipids and cardiovascular risk, as in other patients [Kitas2019], and larger, longer placebo-controlled trials with disease activity as the primary end point are needed before statins can be used to treat RA itself.

4.2.2. Fenofibrate

Fenofibrate is a hypolipidemic drug and a synthetic PPAR-α ligand [Varga2011]. In mouse T cells, fenofibrate (5–20 μM) inhibited Th17-cell

differentiation in a dose-dependent manner, by 74% at 20 μM, whereas other PPAR-α agonists (WY14643, GW7647, and bezafibrate) had no such effect, which suggests that fenofibrate acts here through another pathway [Zhou2012]. In AIA rats, fenofibrate (300 mg/kg/day for 12 days) lowered the arthritis score from 11.0 to 5.83, a fall of about 47% ($P < 0.01$) [Castillero2012]. This dose was far above clinical doses. By the body-surface-area method for a 60-kg adult, 300 mg/kg/day corresponds to approximately 2,900 mg/day, about 15–20 times the 145–200 mg/day used in RA studies [NairJacob2016][Castillero2012][Shirinsky2013][Goto2010]. In a randomized study without a placebo arm, 44 patients with RA and dyslipidemia were assigned to fenofibrate 200 mg/day (n = 23) or statins (n = 21) for 6 months [Goto2010]. At 6 months, the prednisolone dose was lower in the fenofibrate arm than in the statin arm ($P = 0.0449$) [Goto2010]. Within the fenofibrate arm, it had fallen ($P = 0.0098$ vs baseline) from a mean of 3.6 (standard deviation [SD] 2.7) to 2.0 (SD 2.2) mg/day [Goto2010]. However, this small study tested several outcomes, so the result is exploratory. CRP changes were non-significant in both arms [Goto2010]. In an uncontrolled pilot study, 27 patients with active RA received 145 mg/day of fenofibrate for 12 weeks in addition to stable DMARDs, and median DAS28 fell from 6.28 to 4.83 [Shirinsky2013]. Without a control group, part of this fall may reflect regression to the mean in patients enrolled during high disease activity.

CRP and IL-6 decreased by 36% and 41%, respectively, while 2 patients withdrew because of severe back pain considered possibly drug-related [Shirinsky2013]. Clinical evidence therefore remains limited to these two small studies, and to our knowledge no placebo-controlled trial has evaluated fenofibrate in RA.

4.3. Angiotensin II Receptor Blockers

The renin–angiotensin system (RAS) regulates blood pressure and fluid balance, and its overactivity is a common cause of hypertension. The RAS may also contribute to RA, particularly through angiotensin-converting enzyme (ACE). ACE activity released by peripheral blood monocytes was raised in 29 of 32 patients with RA [Goto1990]. In synovium from 7 patients with RA, stromal ACE immunostaining and ACE-inhibitor binding were higher than in osteoarthritis or chondromalacia patellae, whereas vascular binding did not differ [Walsh2000].

In rats with arthritis induced by complete Freund's adjuvant (CFA), protein expression of angiotensin II type 1 receptors (AT1R) was higher and that of angiotensin II type 2 receptors (AT2R) was lower than in normal rats ($P < 0.05$) [Hussein2025]. Conversely, when the AT2R agonist CGP42112 (10 and 20 μg/kg) was injected into the joint of AIA rats, the arthritis index of the treated paw fell on days 22 and 26 [Wang2013].

Animal and cell studies of losartan are largely

consistent. In AIA rats, losartan at 10 and 15 mg/kg/day significantly lowered the polyarthritis index compared with vehicle, although this effect was smaller than that of MTX [Wang2013]. When converted to a 60-kg adult with the Km ratio method, these oral doses correspond to an estimated 97 and 146 mg/day [Wang2013][NairJacob2016]. Losartan also shifted the receptor balance in the spleen and synovium towards AT2R, with less AT1R and more AT2R, a change that paralleled the fall in the polyarthritis index [Wang2013]. In a second study in AIA rats, losartan lowered serum IL-6 and TNF-α compared with the vehicle-treated arthritic group; these changes were accompanied by a rise in glutathione (GSH) and a fall in serum malondialdehyde (MDA), two markers of oxidative stress [AbdelElGaphar2018]. A third study, by a different group, used CFA-induced arthritis in rats and tested oral losartan at 20 mg/kg/day [Hussein2025]. Relative to untreated CFA rats, oral losartan reduced angiotensin II (Ang II) in the joint from 170.59 to 22.41 pg/g, IL-6 from 320.78 to 129.43 pg/g, and TNF-α from 325.80 to 130.99 pg/g [Hussein2025]. In the same study, oral losartan also lowered joint tissue nicotinamide adenine dinucleotide phosphate (NADPH) oxidase from 75.10 to 18.29 ng/g and nitric oxide (NO) from 37.32 to 10.53 μmol/g [Hussein2025]. In one in vitro study of stimulated mononuclear cells from 16 patients with RA, losartan (100 μM) lowered interferon-γ (IFN-γ), IL-6, IL-17F, and IL-22 [Cardoso2018]. However, cytokines were measured at a

single concentration (100 μM), and the response was weaker in more active disease, with only IL-22 still reduced at DAS28 > 3.2 [Cardoso2018]. These results may therefore serve only as supporting evidence.

The main clinical evidence for this class in RA comes from a trial of another angiotensin II receptor blocker (ARB), candesartan. In this Egyptian single-blind, randomized, placebo-controlled trial, 45 patients with RA (15 per group) were treated for 3 months with add-on candesartan (8 mg/day), atorvastatin (20 mg/day), or placebo [Mostafa2020]. Compared with the control group, candesartan significantly lowered serum IL-1β, TNF-α, and CRP and improved DAS28 and the disability index of the Health Assessment Questionnaire (HAQ-DI), whereas ESR was reduced only relative to baseline [Mostafa2020]. Atorvastatin lowered both MDA and rheumatoid factor (RF), but candesartan changed neither significantly [Mostafa2020]. These findings should be interpreted with caution, because patients were grouped according to their other medical conditions, only 45 of 59 eligible patients were analysed, the groups differed at baseline in several variables, including CRP and HAQ-DI, and follow-up was short [Mostafa2020].

ARBs, particularly losartan, may therefore be a reasonable choice for patients with RA who also need treatment for hypertension, although larger and longer trials of losartan are required to establish any disease-modifying effect.

4.4. Emerging Candidates

Other drugs, including propranolol, sertraline and valproate, have been proposed for RA, but the evidence for them is hypothesis-generating or not supportive (Table 2).